Plant Disease Detection and Classification Using Deep Learning Models*

S. Konduru[1], M. Amiruzzaman[1], V. Avina[1], M. R. Islam[2]
[1]Department of Computer Science
West Chester University
West Chester, PA 19383, USA
{sk984965,mamiruzzaman,va985361}@wcupa.edu
[2]Department of Computer Science and Engineering
Bangladesh University of Business and Technology
Rupnagar, Mirpur-2, Dhaka-1216, Bangladesh
md.rajibul.islam@bubt.edu.bd

Abstract

Detection of diseases in plants at an early stage is crucial to achieving high yields, preserving crop quality, and effective disease management. Existing research focuses mostly on leaf disease detection, despite the fact that disease may develop everywhere on the plant. We developed a new dataset using the PlantVillage dataset and other online sources. We used Convolutional Neural Network (CNN) architectures, Alexnet and MobileNet to analyze and evaluate the performance of the models on the new dataset (i.e., consists of over 50,000 images). The models were trained on the new dataset for 100 epochs. MobileNet outperformed the other two models, attaining 99.69% training accuracy, 94.37% validation accuracy, 96% average precision, 96% recall, and an F1-score. The MobileNet model predicted diseases that affect portions of the plant other than the leaf better. This work demonstrates detecting plant disease and provides a feasible technique for enhancing crop management.

1 Introduction

Agriculture is an essential and underrated sector that provides us with our major source of food. Because of the expanding population, there is a need for increased food production. In the last decade, many agricultural fields have been transformed into commercial lands because of urbanization, perhaps leading to food scarcity. Crop production must be increased to meet food requirements over the increasing population. Pests and diseases are two of the most significant elements influencing crop yield [5]. It is critical to diagnose ailments early and treat them effectively to obtain high yields. This also helps to retain the quality and quantity of the food [3].

Other benefits include low-cost treatment, preventing the disease from spreading further, etc. Overall, it is necessary to detect plant diseases for increased food production, food security, and environmental protection [2, 3]. The most common cause of plant diseases is pathogens such as bacteria, fungi, viruses, and nematodes. These pathogens feed on plants and absorb nutrients from the plant, shrinking the growth and further damaging the leaves, stems, and roots, leading to a systematic infection that spreads through the entire plant resulting in death.

Following the guidelines and experience from existing work, the proposed work uses Convolutional Neural Network based algorithms such as MobileNet and AlexNet for the detection and classification of plant diseases. These models seems to have better performance in classifying and detecting plant-based diseases [1, 4, 7]. The work focuses on detecting diseases that occur on other parts of the plant besides leaf. To detect the disease at early stage, it is necessary to diagnose the entire plant, not just the image. The proposed model is intended to consider the whole plant for the diagnosis. Due to the lack of data on other parts of the Apple crop, data augmentation was performed by collecting the images from google. This study only focused on detection and identification of the plant diseases not the source of the diseases and economic aspects.

Three different crops, such as apple, tomato, and corn, are considered for training the model. Tomatoes and corn have only leaf images; we tried to cover most of the other parts of apples. A dataset containing 25 distinct types of diseases (classes) associated with apples, corn, and tomatoes is considered. The images that belong to leaf diseases are extracted from the Plant Village dataset, while the images diseases that occurs on stems, roots and fruits are augmented from the images collected from the web. The training dataset consists of 40,000 images while the testing dataset consists of 10,000 images. The main objective is to analyze the performance of the models on the newly created dataset. Once these models are trained, their performance is compared based on accuracy and latency.

Some of the contributions of the proposed work include:

1. Data balancing using under-sampling and oversampling to balance the class distribution in training data. Various data augmenting techniques are used to generate more images from the limited existing images through geometric and color transformations.

2. The model is trained to detect diseases that occur on all parts of the plant, not just the leaves. There are no datasets available on diseases that occur on stems, fruits, or roots. The current datasets only detect leaf disease, which is why we generated a dataset by gathering and augmenting relevant images from Google and other sources. This dataset contains data that was acquired in an uncontrolled environment.

3. Multiple CNN models are trained and tested to see how they learn and perform on the new dataset.

2 Related work

We studied some related work and they helped us to gain valuable insights and what are the shortcomings in the existing works. A synopsis of the related works are presented below:

In [4], Mahmudul et al. replaced the standard convolution method to reduce the computation cost and the number of parameters. The work focused on leaf diseases in 14 different plant species. Similarly, Sasikala et al. [11] also used different CNN models, such as, ResNet 50 & 101, InceptionV3, DenseNet121 & 201, MobileNet V3, and NasNet. To overcome overfitting of the model, various data augmentation techniques such as image enhancement, scaling, rotation, and translation were applied. The Plant-Village dataset, that consists of around 38 classes belonging to 14 different types of crops was used to train the pretrained models.In [7], Swathi et al. proposed a model called AgriDoc for the detection and classification of plant leaf diseases. The Plant Village dataset was used, which consists of 38 different classes. Different convolutional neural network models, such as AlexNet, MobileNet, ResNet and some built from scratch CNN-based models were implemented and compared based on their performance and accuracy.

In [1], Rinu et al. used VGG16, which is a CNN based model for the detection of plant leaf diseases belonging to 38 classes. In [2], Chowdhary et al. achieved 98% accuracy on average using U-net models. Furthermore, the authors achieved 99% accuracy as they used EfficientNet-B4 model. The work showed that the proposed models performed better when they were trained on segmented images with deeper networks. It was mentioned that the proposed work outperformed the existing work for the same purpose. Not as high accuracy as [1] and [2]. But, Nishant et al. [10] achieved 95.6% accuracy based on a CNN-based model (i.e., VGG19) and detected and classified plant leaf diseases that belong to 13 different species. Arun Pandian et.al [9] achieved 99.79% accuracy on average. In [8], Lili Li et al. reviewed different works on plant disease detection and classification using deep learning techniques in the past year.

In [3], Haridasan et al. proposed a system that detects rice crop diseases using computer vision, image processing, machine learning, and deep learning techniques. The authors aimed to achieve efficiency and reduce losses in the farming industry. Some of the most common rice diseases, such as brown leaf spot, rice blast, sheath rot, bacterial leaf blight, and false smut, are considered for training the model. Advanced techniques like image segmentation, support vector machine classifier, and convolutional neural network algorithms have been employed to detect and classify diseases accurately. The CNN method achieved better results than SVM in terms of accuracy. The model achieved 91.45% accuracy with ReLu and Softmax activation functions.

After studying and analysing related works, we can say that the existing works focused only on detecting leaf diseases while ignoring that disease can occur on any part of the plant, including root, stem, fruit, etc. Most of the existing works used the Plant Village dataset, which has a class imbalance problem. Some of the classes in the dataset have a greater number of images, while other classes have fewer images. If there is a huge imbalance between the classes, the model prioritizes in learning more from majority class thus resulting in huge error rate in predicting from minority class. Model will be biased and over-predicts the majority class and under-predicts the minority class further resulting in poor accuracy. Existing works have trained models on datasets with limited scope of diseases.

3 Methodology

The methodology for plant disease detection involves multiple phases that include (see Figure 1):

3.1 Data Collection

The images that belong to Apple, Corn and Tomato leaf diseases are acquired from the existing dataset, Plant Village while the images that belong to the diseases that occur on other plant parts are acquired from google. Due to the lack of images related to certain diseases, only 50-100 images that belong to 9 different classes were downloaded and labelled properly. The data from these 9 classes were balanced through oversampling using data augmentation

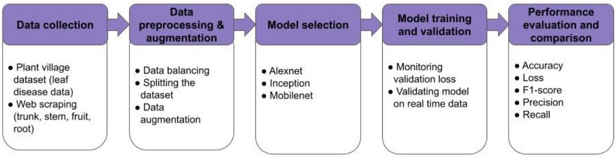

Figure 1: Overview of Proposed Methodology

techniques which will be discussed in detail in the further section. The images that belong to 16 classes were balanced through under-sampling by using splitting function. Overall, the dataset contains 50,000 images out of which 40,000 images were used for training the model and 10,000 images were used for validating the model. Out of 25 classes, each class contains 2000 images before they were split into training and testing datasets.

3.2 Data Pre-processing and Augmentation

The images that are downloaded from google are carefully examined and the images that have a lot of noise are removed. Once the images that belong to different classes are properly selected, they are further augmented using different augmentation techniques (see Figure 2):

Figure 2: Illustration of different transformations applied to the original image.

Data augmentation is a method used to increase the size and diversity of the existing data by applying either geometric or color transformations. Geometric transformations include flipping, cropping, rotating (0 to 360), zooming and scaling of an image to generate new images while the color transformations include brightness and contrast. These transformations help the model to generalize unseen data and reduce over-fitting.

3.3 Model Selection

The main focus of the proposed work is not to achieve high accuracy using a particular deep learning model; it is to analyze how the deep learning models are performing on the augmented data that belongs to the whole plant and not just leaf. In this work, AlexNet and MobileNet models are chosen as they were proven to have achieved standard and consistent results in many state of the art approaches. Also, MobileNet consumes less computational power and less execution time.

3.3.1 AlexNet

AlexNet is one of the signifant convolutional neural networks developed in 2012 [6], by Alex krizhevsky, Ilya Sutskever and Geoffrey Hinton. This architecture consists of 8 layers in which the first 5 layers are convolutional layers and the remaining three are fully connected layers.

3.3.2 MobileNet

MobileNet is also a convolutional neural network which is known for its ease of computation compared to other CNN architectures. It contains multiple convolutional layers that are followed by average pooling layer and a SoftMax output layer that is fully connected. Image pre-processing and normalization is performed in the first layer. This architecture consists of two different convolutions such as depthwise and pointwise convolution.

3.4 Experimental Setup

Deep learning models typically require significant computational resources to train, especially for larger and more complex models or datasets. The dataset that is used for training has over 50,000 images.

3.5 Model Training

The training and validation results for each model is discussed below.

3.5.1 AlexNet

The model was trained on augmented dataset for 100 epochs with batch size 32 monitoring the loss. The model ran for 30 hours continuously.

The model achieved 99.19% training accuracy and 91.58% validation accuracy at epoch 100 whereas the training loss was 0.0286 and the validation loss was 2.6838 at epoch 100. The accuracy with the best loss value is 88.70%

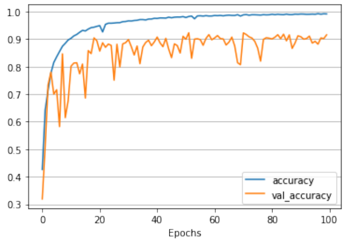

Figure 3: Training vs Validation accuracy of AlexNet model

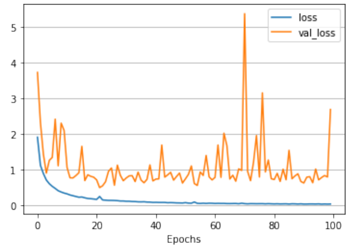

Figure 4: Training vs Validation loss of AlexNet model

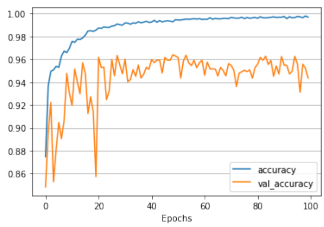

Figure 5: Training vs Validation accuracy of MobileNet model

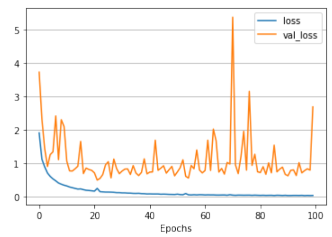

Figure 6: Training vs Validation loss of MobileNet model

(see Figure 4). The model is evaluated in terms of accuracy, F1 score, loss, precision, and recall.The trained model weights are saved as a h5 file for future use and the model data is saved to CSV format for data analysis.

3.5.2 MobileNet

The model was trained on augmented dataset for 100 epochs with batch size 32. Pre-trained imagenet weights were used and the layers were not re-trained on the new dataset. The model was initially trained by retraining some of the layer on the augmented dataset, but the results showed overfitting because of which we kept experimenting with layers retraining but the results were no so prominent either of the cases. Finally, we tried to keep the Imagenet weights as it is without retraining the layers and the results looked better as well. The accuaracy and the loss can be seen in (5) and 6).

The evaluation is performed in terms of accuracy, precision, loss and recall.

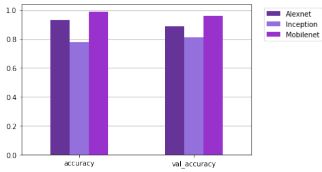

Figure 7: Comparing training & validation accuracy of the models

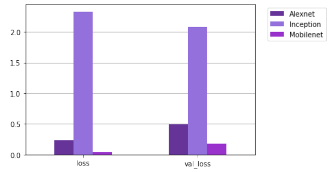

Figure 8: Comparing training & validation loss of the models

3.6 Performance Evaluation

The performance of the models AlexNet and MobileNet are compared in this section in terms of accuracy, loss, precision and recall.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

$$F1\ Score = 2 \times \frac{(precision \times recall)}{(precision + recall)} \tag{4}$$

TP represents true positives, TN represents true negatives, FP represents false positives, and FN represents false negatives.

The training and validation loss of the models are compared and the results are shown in the Figure 8).The loss of AlexNet is better but it is still high compared to MobileNet. Overall, Mobilenet performed better with lower loss compared to other two models.

Table 1 shows the comparison of models using performance metrics such as accuracy, precision, recall and F1-score. The accuracy values mentioned in Table 1 are the values taken for best validation loss value. The results clearly show the superiority of MobileNet architecture over the other two models. Also, MobileNet took less time for training compared to other two models because of the less number of parameters. The results are not compared with the state of the art approaches as they are trained on data that only has leaf disease

Table 1: Performance Evaluation of CNN models

Model	Accuracy	Precision	Recall	F1 Score
AlexNet	88.70%	0.89	0.89	0.89
MobileNet	95.99%	0.96	0.96	0.96

images. Since this work focuses on diseases that occur on all parts of the plant, it is not efficient to compare the results with the existing methods.

4 Conclusion

Deep learning is one of the advanced technologies that is being implemented in various domains, including agriculture. In this study, we worked on disease detection. In addition, we have created our dataset by collecting images from various sources and augmenting the images to make sure all the classes are balanced and has over 50,000 images that belong to 25 different classes. Only three crops, including apples, corn, and tomatoes are considered for the simplicity purpose.

Convolutional neural network architectures such as AlexNet and MobileNet are considered after thorough research of the existing works. These models are trained on the proposed dataset to analyze and evaluated in terms of accuracy, precision, recall, and F1 score. As per the results, MobileNet performed better than AlexNet model. Also, it was faster in training and predicting disease detection.

As for the future study, real-time photos and augmented images can be added to the dataset to increase the model's performance. Also, using an image translation method, the Cycle GAN architecture may be utilized to expand the dataset.

References

[1] SP Adarsha et al. “Identification of Plant Diseases Based on Lesion Spots”. In: *Journal homepage: www. ijrpr. com ISSN* 2582 (2022), p. 7421.

[2] Muhammad EH Chowdhury et al. “Automatic and reliable leaf disease detection using deep learning techniques”. In: *AgriEngineering* 3.2 (2021), pp. 294–312.

[3] Amritha Haridasan, Jeena Thomas, and Ebin Deni Raj. “Deep learning system for paddy plant disease detection and classification”. In: *Environmental Monitoring and Assessment* 195.1 (2023), p. 120.

[4] Sk Mahmudul Hassan et al. “Identification of plant-leaf diseases using CNN and transfer-learning approach”. In: *Electronics* 10.12 (2021), p. 1388.

[5] Md Rajibul Islam et al. “An Efficient Technique for Recognizing Tomato Leaf Disease Based on the Most Effective Deep CNN Hyperparameters”. In: *Annals of Emerging Technologies in Computing (AETiC)* 7.1 (2023).

[6] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. “ImageNet Classification with Deep Convolutional Neural Networks”. In: *Advances in Neural Information Processing Systems 25 (NIPS’2012)*. 2012.

[7] H Kushal et al. “AgriDoc: Classification and Prediction of plant leaf diseases”. In: *Turkish Journal of Computer and Mathematics Education (TURCOMAT)* 12.14 (2021), pp. 3909–3932.

[8] Lili Li, Shujuan Zhang, and Bin Wang. “Plant disease detection and classification by deep learning—a review”. In: *IEEE Access* 9 (2021), pp. 56683–56698.

[9] J Arun Pandian et al. “Plant disease detection using deep convolutional neural network”. In: *Applied Sciences* 12.14 (2022), p. 6982.

[10] Nishant Shelar et al. “Plant Disease Detection Using Cnn”. In: *ITM Web of Conferences*. Vol. 44. EDP Sciences. 2022, p. 03049.

[11] Sasikala Vallabhajosyula, Venkatramaphanikumar Sistla, and Venkata Krishna Kishore Kolli. “Transfer learning-based deep ensemble neural network for plant leaf disease detection”. In: *Journal of Plant Diseases and Protection* 129.3 (2022), pp. 545–558.